

Preregistration v2.0: Mapping the high- $|z|$ threshold event structure of de Bruijn–Newman heat flows

Date: 2026-10-03 · Author: Abby Davis (independent researcher, Tucson AZ; ORCID 0009-0002-6758-2263), with Melody/Muse as instrument Status: WRITTEN BEFORE ANY v2 $t \neq 0$ COMPUTATION. No flow results for this preregistration have been observed. v1.0's data and code exist and are used ONLY for calibration design, never as v2 evidence. Relationship to v1.0: This is a NEW preregistration, not a continuation or rescue of the 2026-10-02 heat-flow milestone (verdict: METHOD INVALID). It is written to be executable by an independent analyst.

1. What v1.0 established (and what it forbids)

v1.0 (dir: riemann-heatflow/, verdict METHOD INVALID per its own §6) established:

- (a) The preregistered premise " t decreasing $\Rightarrow$ lowest real zero pair approaches and collides" is FALSE for the verified implementation $H_t(z) = \sum c_j e^{\{tu_j^2\}} \cos(u_j z)$ (PDE residual $\leq 3.5e-22$ — the code is correct; the premise was wrong). Low zeros SPREAD as t decreases.
- (b) A post-hoc wide scan found $\sim 20+$ nonreal zeros at HIGH $|z| \approx 190-230$ for $t < 0$, consistent with the rigorous bound $\Lambda \geq 0$ (Rodgers–Tao 2020). The threshold dynamics live at high $|z|$, not the lowest pair.
- (c) Verification gates V1/V2/V4/V5 were never run or logged in v1.0; synthetic controls saved no trajectories; the Davenport–Heilbronn off-axis pair was never located.
- (d) One high- z complex zero appeared t -stationary ("frozen") — possibly a quadrature artifact; unresolved.

v1.0 FORBIDS: re-pointing at high- z zeros inside the v1.0 milestone (moving the goalposts); any discriminator claim from v1.0 data; assuming a collision direction anywhere in this design.

2. Research question and scope

Question. Under the de Bruijn–Newman heat flow, what is the real $\leftrightarrow$ complex transition event structure in a WIDE zero window ($|z| \leq 300$) — and does any of its preregistered features distinguish the ζ flow from RH-false flows when produced by a validated, artifact-quantified

apparatus?

Scope (hard ceiling). Methods/phenomenology paper: "the high-zero threshold event structure of de Bruijn–Newman-type heat flows." NOT a proof of RH, NOT a statement about RH's truth value, NOT a measurement of Λ . A null (no discriminator found) is a complete lab-notebook result and is not published.

3. Conventions

Wide frame (Rodgers–Tao / Polymath): $s = (1+iz)/2$. ζ : $H_0(z) = \xi(1/2+iz)/8$, zeros at $z = \pm 2\gamma_n$. Heat flow: $H_t(z) = \int_0^\infty \Phi(u) e^{\{tu^2\}} \cos(zu) du$, satisfying $\partial_t H_t = -\partial_{zz} H_t$. de Bruijn–Newman constant Λ : H_t is real-rooted $\iff t \geq \Lambda$; proved $0 \leq \Lambda \leq 0.2$ (lower: Rodgers–Tao 2020; upper: Platt–Trudgian 2021).

Event taxonomy (fixed in advance).

- Type-R: two adjacent real zeros collide (gap $\rightarrow 0$); below the collision a complex-conjugate pair emerges. Detected wherever it occurs — no direction assumed.
- Type-C: a complex-conjugate pair's $|\text{Im}| \rightarrow 0$ at a common real point (lands on $\mathbb{R}$).
- Type-X: collision of zeros off the real axis that are not conjugates (codimension 2; expected never; flagged, never fitted).
- ARTIFACT: a detector trigger failing the genuineness protocol (§6). Counted and reported, never mapped.

4. Cases

(a) ζ (reference). H_t via fixed Gauss–Legendre quadrature, $J=256$ nodes on $[0,3]$, `mpmath` `dps=50`, $\Phi(u) = \sum_{n=1}^8 (2\pi^2 n^4 e^{\{9u\}} - 3\pi n^2 e^{\{5u\}}) e^{\{-\pi n^2 e^{\{4u\}}\}}$. (This corrects v1.0's `umax=1.0` deviation: the interval is $[0,3]$ and is gated.)

(b) Synth- $\delta 1$ ("gross" RH-false, ground truth by construction). $H_0^{\text{syn}}(z) = H_0(z) \cdot R(z)$ with $R(z) = [(1-z^2/w^2)(1-z^2/\bar{w}^2)]/(1-z^2/z_1^2)$, $w = z_1 + i \cdot 1.0$. Surgically replaces the real pair $\pm z_1$ with the quadruplet $\pm w, \pm \bar{w}$; R is real on $\mathbb{R}$, $R(0)=1$, $R \rightarrow 1$ at ∞ , and $H_0 \cdot R$ is entire (denominator pole cancels against H_0 's simple zero). $N_{\text{off}}(0) = 4$ by construction. $H_t^{\text{syn}} = e^{\{-t\partial_{zz}\}}[H_0 \cdot R]$ via Taylor-in- t with Leibniz rule, truncation $M=48$ (gated against $M=96$).

(c) Synth-50.1 ("near-miss" RH-false, sensitivity probe). Same as (b) with $\delta = 0.1$ (Rodgers-Tao near-miss scale).

(d) Davenport-Heilbronn (authentic RH-false Dirichlet series). BEST-EFFORT, GATED.

Wide-frame $\Phi_{\text{DH}}(v) = 8e^{\{3v\}} \sum_{n=1}^{\infty} n \cdot a_n e^{\{-\pi n^2 v/5\}}$, $a = (1, \kappa, -\kappa, -1, 0)$ period 5, $\kappa = (\sqrt{(10-2\sqrt{5})-2})/(\sqrt{5}-1)$ (expected 0.284079043840...). Gates (ALL must pass): G1 functional equation $|\Lambda_{\text{DH}}(s) - \Lambda_{\text{DH}}(1-s)| < 10^{-20}$ at 5 random s ; G2 reality $|\text{Im} \Lambda_{\text{DH}}(1/2+it)| < 10^{-20}$ at 5 random t ; G3 quadrature cross-check $|\int \Phi_{\text{DH}} \cos - \Lambda_{\text{DH}}((1+iz)/2)| < 10^{-25}$ (up to one fitted overall scale) at 5 random z . Time-box: 45 minutes wall-clock. On timeout or gate failure: DH is dropped, the drop is logged as a red flag, and the comparison proceeds as ζ vs (b), (c), (e).

(e) Null-window detector check. The ζ up-leg interval $(0.2, 0.5]$ is RIGOROUSLY event-free (real-rootedness is monotone in t for $t \geq \Lambda$). Running the full event detector there measures its false-positive rate on real data under a theorem-grade null — this plays the role the Poisson control played in the anti-resonance study. Any detected event $\rightarrow$ detector bug $\rightarrow$ INVALID (apparatus).

5. Tracking protocol (identical across cases unless noted)

- Window: every zero with $|z| \leq 300$ at $t=0$ ($\approx 60+$ for ζ : $\pm 2\gamma_1 \dots \pm 2\gamma_{30}$; synth cases: $\pm w, \pm \bar{w}$ plus ζ -like rest; DH: as located, or dropped per §4).
- $t=0$ seeding: grid-seeded complex Newton (tol $|\Delta z| < 10^{-30}$, max 25 iters, require $|\text{H}_t(z)| < 10^{-25}$ at convergence), deduplicated by pairwise distance.
- Legs: down-leg $t \in [0, -1.0]$; up-leg $t \in [0, 0.5]$ (synthetic landing checks; one preregistered extension to 2.0 per §8(iv); DH off-axis pair up to 1.0 if gated in).
- Step control: $\Delta t = 0.01$ initial; halve if Newton iters > 12 (floor 0.001); double if iters < 4 (cap 0.02); reject any step with $|\Delta z| > 0.5 \times \text{neighboring gap}$, halve Δt and retry.
- Detection: monitor all adjacent real gaps $g_k(t)$ and all complex pairs' $|\text{Im } z_k(t)|$. Trigger R when $g_k < 0.03$ (log (t, g_k^2) each step; at $g_k < 10^{-7}$ or Newton failure, fit $g^2 = C^2(t-t^*)$ on points with $g \in [10^{-7}, 3 \times 10^{-2}]$, require ≥ 5 points and $R^2 \geq 0.999$). Trigger C when $|\text{Im}| < 0.03$ (fit im^2 linearly under the same quality rules). Trigger X is logged with coordinates and never fitted.
- Trajectories: FULL JSON per case — every t -step, every tracked zero ($z, |H|$, Newton iters, Δt). v1.0 failed here; in v2 this is mandatory. A case without saved trajectories is VOID.

6. Per-event genuineness protocol (answers v1.0's frozen-zero problem)

Every triggered event must pass ALL of the following, or it is an ARTIFACT:

- G-a (t-continuation): the participating zeros track smoothly (per-step $|\Delta z|$ within the §5 step-rejection bounds) for ≥ 10 steps on each side of the event. Non-smoothness $\rightarrow$ SUSPECT: logged, excluded from the map.
- G-b (J-refinement): recompute the event at $J=512$; require $|\Delta t^*| < 3\sigma(\text{fit})$ and $|\Delta z^*| < 0.05$. Failure $\rightarrow$ ARTIFACT.
- G-c (PDE): $|\partial_t H + \partial_{\{zz\}} H| < 10^{-20}$ at the fitted (z^*, t^*) .

Artifact failures are logged with reasons and tallied; the artifact rate $A/(A+|E|)$ is a reported observable. If it exceeds 0.5 for ζ , the apparatus is declared UNFIT for high- z work $\rightarrow$ INVALID (apparatus). That is a verdict about the instrument, not the theory.

7. Verification gates (ALL must pass AND be logged before Phase 2 begins)

- V1 (quadrature): $J=256 \rightarrow J=512$ changes $H_t(z)$ by $< 10^{-35}$ at 10 test (z,t) , INCLUDING 4 points with $|z| \in [180, 280]$. (v1.0 never tested high z .)
- V2 ($t=0$ zeros): $|H_0(2\gamma_n)| < 10^{-30}$ for $n = 1\dots 5$; 14-digit agreement with $2\gamma_n$.
- V3 (PDE): $|\partial_t H + \partial_{\{zz\}} H| < 10^{-20}$ at 10 test (z,t) per case. (v1.0 passed this unlogged; re-run and LOG it.)
- V4 (synthetic): $M=48$ vs $M=96$ agree to $< 10^{-25}$ at all tracked (z,t) ; $H_0^{\text{syn}}(w) = 0$ to $< 10^{-25}$ (construction check); PDE residual $< 10^{-20}$.
- V5 (seed independence): the full ζ down-leg is repeated with Newton seeds perturbed by 10^{-6} ; the VALIDATED event map must agree — same events within 3σ , none gained, none lost.
- V6 (detector null): the full event detector is run on ζ over $(0.2, 0.5]$; zero events required.

Any gate failure $\rightarrow$ STOP, diagnose, red-flag. Phase 2 does not begin until every gate passes and its log exists on disk.

8. Calibration (ζ + synth must satisfy ALL, else METHOD INVALID)

- (i) $N_{\text{off}}(0) = 0$ for ζ in $|z| \leq 300$.
- (ii) V1–V6 all pass and are logged.
- (iii) Zero validated events in ζ over $(0.2, 0.5]$ (same computation as V6; restated here as the flow-level check).
- (iv) APPARATUS VALIDATION — the rigorous direction: synth-61's injected pair lands via Type-C at t^*_{inj} with $0 < t^*_{\text{inj}} \leq 2.0$. The up-leg is tracked to 0.5 and extended ONCE to 2.0 if and only if no landing occurred by 0.5 (preregistered extension, not peeking). The landing requires an im^2 -fit with $R^2 \geq 0.999$, and N_{off} must drop by exactly the injected count at the landing. If no landing by $t = 2.0 \rightarrow \text{INVALID}$.

Item (iv) replaces v1.0's dead calibration (ii) with a claim the theorems actually support: the heat flow real-ifies as t increases (de Bruijn). v1.0 calibrated on the direction the flow does NOT go; v2 calibrates on the direction it provably does.

9. Phases, observables, decision rule

Phase 1 (characterization, ζ only). Produce the validated event map E_{ζ} and the artifact tally A_{ζ} . This phase is descriptive; it makes no discriminator claim. If E_{ζ} is EMPTY (no genuine events survive G-a/b/c), report that plainly — Phase 2 still runs, and the comparison becomes "empty vs non-empty maps."

Phase 2 (comparison). The identical protocol on cases (b), (c), (e); case (d) if gated in.

Observables. E = validated event map $\{(\text{type}, t^* \pm \sigma, z^*)\}$; A = artifact count and rate; $N_{\text{off}}(t)$ curves per case; t^*_{inj} for the synthetic cases.

Decision rule (fixed now).

- SUCCESS: calibration (§8) passes AND at least one preregistered feature separates ζ from at least one RH-false case at $>3\sigma$. Preregistered features: $|E|$ (event count); the distribution of $|z^*|$ (two-sample Kolmogorov–Smirnov, $\alpha = 0.01$ — if either map has fewer than 5 events this feature scores as non-separating for lack of data, never as a success); t^*_{max} (latest event time); $N_{\text{off}}(-1)$. Stability of the separating feature across the V5 seed check is required.
- FAILURE: calibration passes but no feature separates (all within 3σ) — INCLUDING the sub-case of an empty E_{ζ} .

- INVALID: any calibration item fails, any gate fails, or the ζ artifact rate exceeds 0.5.

Interpretation cap. SUCCESS means: "the high-z heat-flow event structure is a validated dynamical discriminator between RH-true and RH-false flows on the tested cases." It does NOT imply anything about the truth value of RH. It is NOT a measurement of Λ (all t^* values are windowed).

10. Compute boxes

- DH gating: 45 minutes wall-clock (as v1.0).
- ζ wide-window down-leg: 6 hours wall-clock; on timeout, report the partial map with the box noted — a preregistered scope cut, not a failure.
- Synthetic cases: 3 hours each.
- Total study: 48 hours wall-clock from the first v2 $t \neq 0$ evaluation.

11. Deliverables

- code/: `ht_flow.py` (quadrature core + H_t and derivatives), `track.py` (wide-window Newton tracker + event detector + genuineness protocol), `synthetic.py`, `dh.py` (gated), `analyze.py`, `make_figures.py`.
- data/: per-case trajectory JSONs (mandatory), event table CSV including genuineness columns, gate-check log (mandatory), artifact log.
- figures/: (1) ζ event map (t vs z , type-encoded); (2) fold/landing fits with residuals; (3) $N_{\text{off}}(t)$ for all cases; (4) artifact gallery (triggers that failed G-a/b/c, with reasons).
- results.md: quantitative outcomes, the full red-flag log, and a soundness self-assessment.
- This preregistration, with its sha256 recorded in results.md.

12. Pre-declared red flags and known risks

- R1 (v1.0 R-A): never assume a collision direction again. The protocol detects R/C/X wherever they occur; the map is the datum.
- R2 (v1.0 R-B): frozen / quadrature-stationary zeros at high z . G-a/b/c exist for exactly this; artifact rate $> 0.5 \rightarrow$ INVALID (apparatus).

- R3 (v1.0 R-C): the DH off-axis pair may again evade location → gated drop, logged, comparison proceeds without it.
- R4 (v1.0 R-D): synthetic trajectories are MANDATORY; a case without saved trajectories is VOID, not "log-only."
- R5 (v1.0 R-E): $u_{\max}/J/dps$ are as specified in §4; any deviation is a protocol deviation — logged, and the affected gates re-run.
- R6: Newton degradation near folds ($H' \rightarrow 0$ at the fold): fit-early strategy as v1.0; never Newton into $g < 10^{-7}$.
- R7: label jumping between adjacent zeros at small gaps: step rejection plus a post-hoc trajectory-continuity check; jumps trigger re-tracking or a flag.
- R8: compute cost at $|z| \sim 300$ ($dps=50$, $J=256/512$) may force J/dps tradeoffs; any change is a deviation — logged and re-gated per R5.
- R9 (v1.0 R5, "the boring reason"): if ONLY synth- $\delta 1$ separates, the claim is weakened to "detects gross violations" and reported as such. The $\delta=0.1$ case is the sensitivity probe that earns a stronger claim.
- R10: the t^*_{inj} extension ($0.5 \rightarrow 2.0$) is preregistered SOLELY for calibration item (iv). It is not a general license to extend windows.

End of preregistration v2.0. No v2 $t \neq 0$ computation may precede the sha256 timestamp recorded in results.md.